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Regular Article

Effect of L-theanine on sensorimotor gating in healthy human subjects

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Aim: L-Theanine (N-ethyl-L-glutamine) is an amino acid uniquely found in green tea. Growing evidence has suggested the possible effects of L-theanine on cognition. Previously, we found that L-theanine attenuates MK-801-induced deficit in prepulse inhibition (PPI) in mice. In this study, we examined the effect of L-theanine in increasing the PPI in healthy humans.

Methods: The subjects were 14 healthy adults who underwent PPI testing as a measure of sensorimotor gating 90 min after an oral intake of L-theanine (0, 200, 400, or 600 mg). PPI tests were done by examiners who were blind to the dose.

Results: The administration of 200 mg of L-theanine and that of 400 mg, but not 600 mg, significantly increased the % PPI compared to the baseline (0 mg). There was no significant relation between the dose of L-theanine and the startle magnitude or the habituation of startle response. The plasma concentrations of L-theanine correlated with the dose of L-theanine.

Conclusion: The observed effect with 200–400 mg of L-theanine on PPI suggested that L-theanine at a particular dose range increases sensorimotor gating in humans.

Key words: glutamine, L-theanine (*N*-ethyl-L-glutamine), N-methyl-D-aspartate, prepulse inhibition, sensorimotor gating.

L-THEANINE (*N*-ETHYL-L-GLUTAMINE) was originally found in green tea. Theanine accounts for approximately 50% of the total amino acids in green tea leaves. It comprises about 1–2% of the total dry weight of the green tea leaves, and the median amount of L-theanine per cup of green tea is 8–30 mg.[1]

L-Theanine's chemical structure is similar to that of glutamate, it affects glutamatergic neurotransmission,[2–4] and some monoaminergic neurotransmissions, such as serotonin and dopamine.[5–9] Some previous studies showed that L-theanine has an influence on cognition and mood.[2,10,11] Animal studies showed that L-theanine has an influence on cognition in the rat.[2,9] They revealed the potential cognitive enhancing effects of L-theanine, particularly concerning learning and memory. Regarding human studies, however, there have only been a few studies focusing on cognition, and these results were inconsistent about the relation between the administration of L-theanine and cognition.[12–15]

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The startle response can be suppressed when the startling stimulus is preceded by a weak non-startling prepulse, in a process called prepulse inhibition (PPI). The degree to which such a prepulse inhibits the startle reflex in PPI is used as a measure of sensorimotor gating, by which excess or trivial stimuli are screened or 'gated out' of awareness, so that an individual can focus attention on the most salient aspects of the stimulus-laden environment.[16] Deficient PPI is observed in several neuropsychiatric disorders characterized by abnormalities in the cortico-striato-thalamic-pontine circuitry. Particularly in schizophrenia, PPI was proposed as a measure of information-processing abnormalities.[17–19] PPI is varied following treatment with the N-methyl-D-aspartate (NMDA) receptor antagonist, amantadine, ketamine, memantine, MK-801, and phencyclidine (PCP), and these points indicate that PPI is regulated not only by dopamine inducer, as is well known, but also by glutamatergic neurotransmission.[20–25] However, the PPI changes treated with NMDA receptor antagonists appear to be variable depending on timing and administrated dose.[20–24] In our schizophrenic patients, we obtained evidence for reduced PPI; this was the first evidence of reduced PPI in an Asian population.[26] We subsequently demonstrated that the administration of L-theanine significantly attenuated MK801-induced deficits of PPI in mice.[25] However, to the best of our knowledge, no study has assessed the effect of L-theanine on PPI in humans, and it is necessary to investigate the optimum dose of L-theanine for increase of PPI. To start, in the present study, we investigated whether L-theanine modifies PPI in healthy adults.

METHODS

Subjects

The subjects were 14 healthy adults, including seven men and seven women, whose mean age was 31.0 ± 7.0 years and mean body mass index was 22.5 ± 2.8. Two out of the 14 subjects smoked at least 10 cigarettes per day. The participants were interviewed for enrollment using the Japanese version of the Mini-International Neuropsychiatric Interview (MINI) by a research psychiatrist (M.O., H.H., or T.T.), and only those subjects who had no history of psychiatric illness or contact with psychiatric services were enrolled.[27,28] Participants were excluded if they had a prior medical history of central nervous system disease or severe head injury. Written informed consent was obtained for participation in the study from all subjects, and the study was approved by the Ethics Committee of the National Center of Neurology and Psychiatry, Japan.

Administration of L-theanine

L-Theanine was given to each participant four times, at 0, 200, 400, and 600 mg in a randomized order. Using tablets containing 0 mg (placebo) or 50 mg of L-theanine (Taiyo Kagaku, Yokkaichi, Japan), the subjects were handed 12 tablets composed of L-theanine and/or placebo tablets at each of the four administration times. The half-life of L-theanine ranges from 58 min to 74 min in humans. The washout period takes at least 2 days.[29] In two rat studies, the serum L-theanine concentration reached a peak 0.5 to 2 h after the oral administration of L-theanine (200 mg/kg),[30] and orally administered L-theanine entered the brain through the blood–brain barrier and became detectable 1.5 h post-administration.[7] In humans, the effect of L-theanine on brain activity began to appear 0.5 h after oral administration.[31,32] Based on these findings, our participants were administered L-theanine 1.5 h prior to the PPI test.

Venous blood was sampled to obtain the plasma level of L-theanine immediately before the PPI test. The plasma concentration of L-theanine was calculated using the AccQ-Tag method, according to the manual described in the AccQ-Tag Chemistry Package (Waters Corp., Milford, MA, USA).

PPI test

We used the equipment, setup, and standard PPI test procedures as described,[26,31] but modified and shortened the test procedure. Startle reflex of the subjects to acoustic stimuli was measured using the Startle Reflex Test Unit for Humans (O'Hara Medical Co., Tokyo, Japan). Subjects refrained from smoking for at least 20 min prior to the test. Broadband white noise (50 to 24 000 Hz at 70 dB) was presented as the background noise and was sustained throughout the session. Acoustic startle stimuli of the broadband white noise were presented through headphones. During the initial 3 min of each session, the background noise alone was given for acclimation. In total, 35 trials of startle response were recorded in a session.

The trials consisted of three blocks. In the first block, startle response of the subjects to a pulse

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(sound pressure 115 dB; duration 40 ms) alone was recorded for five times. In the second block, the startle response to the same pulse with or without a prepulse (sound pressure 86 or 90 dB; duration 20 ms; lead interval [onset to onset] 60 or 120 ms) was measured five times for each condition. The differing trial conditions of trials were presented in a random order. In the final block, startle response to pulse alone was again measured for five times. The inter-trial intervals (15 s on average, range 10–20 s) were randomly changed. The entire session lasted approximately 15 min. The analysis of PPI of the subjects was based on the mean of the five trials for each trial type.

We obtained measures of (i) startle response to pulse-alone trials in the first block (digital unit); (ii) habituation (%) of the startle response during the session calculated by the formula ([1 – mean startle magnitude in block 3/mean startle magnitude in block 1] × 100); and (iii) %PPI, calculated by the formula ([1 – mean startle response with prepulse trials)/mean startle response to pulse-alone trials in the second block] × 100). PPI tests were done by examiners who were blind to the dose of L-theanine.

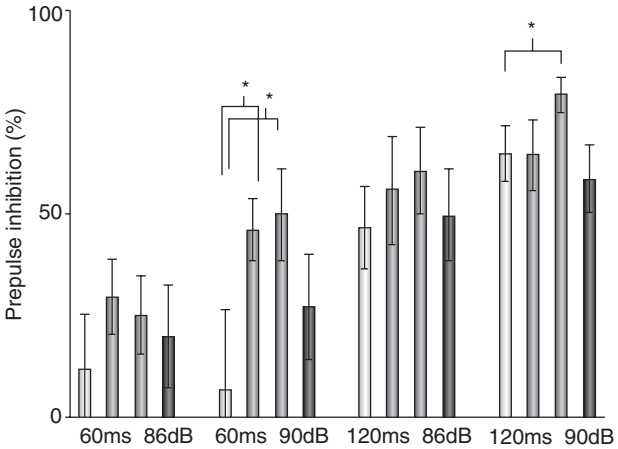

Figure 1. The effects of L-theanine on prepulse inhibition. Repeated measures ANCOVA revealed significant differences in the mean % PPI with the L-theanine administration and the conditions of 60 ms lead interval and sound pressure 90 dB, and of 120 ms and 90 dB. Error bar = SE. $*P < 0.05$ (corrected). □, Baseline; ▨, 200 mg; ▨, 400 mg; ■, 600 mg (Oral administration).

Statistical analysis

To examine the effect of L-theanine on the % PPI, startle magnitude, and habituation, we ran repeated measures Quade's rank ANCOVA for each metric with lead intervals, sound pressure, and dose of L-theanine as within-subjects factors, sex as a between-subjects factor, and age and bodyweight as covariates, followed by Bonferroni-corrected pairwise comparisons as post-hoc tests. A two-tailed P-value < 0.05 was regarded as significant. We evaluated the effect of L-theanine on the startle magnitude and habituation with repeated measures ANCOVA with sex as a between-subjects factor, and age and bodyweight as covariates.

We calculated the association between the plasma concentration of L-theanine and PPI metrics of subjects by a partial correlation analysis with age as a control variable. We also examined the association between the dose of oral administration and the plasma concentration of L-theanine, using Spearman's correlation test. Analyses were conducted with SPSS ver. 20 (SPSS Japan, Tokyo, Japan).

RESULTS

Figure 1 shows the % PPI values depending on lead interval for the prepulse, sound pressure of the prepulse, and dose of L-theanine. There was a significant effect of sound pressure of prepulse on the % PPI values (F = 26.2, d.f. = 1, $P < 0.001$), and of the lead interval (F = 14.8, d.f. = 1, $P = 0.002$). There was a significant interaction between the dose of L-theanine and sound pressure (F = 3.0 d.f. = 3, $P = 0.044$) in the effect on % PPI. However, there was no significant interaction between the dose of L-theanine and lead interval (F = 0.7, d.f. = 3, $P = 0.55$). There was also no significant interaction between sex and lead interval (F = 0.2, d.f. = 1, $P = 0.69$) or between sex and sound pressure (F = 0.8, d.f. = 1, $P = 0.39$).

When the prepulse sound pressure was 90 dB, the % PPI was significantly increased by L-theanine. When the lead interval was 60 ms, there was a significant difference in the % PPI between the baseline and the 200 mg administration ($P = 0.048$) and between the baseline and the 400 mg administration ($P = 0.014$). With the conditions of 120 ms and 90 dB, there was a significant difference in % PPI between the baseline and the 400 mg administration ($P = 0.033$). As shown in Figure 1, a non-linear relation between the dose of L-theanine and % PPI was found.

No other significant differences were found in startle magnitude or the habituation of startle response depending on the dosage of L-theanine administration (data not shown).

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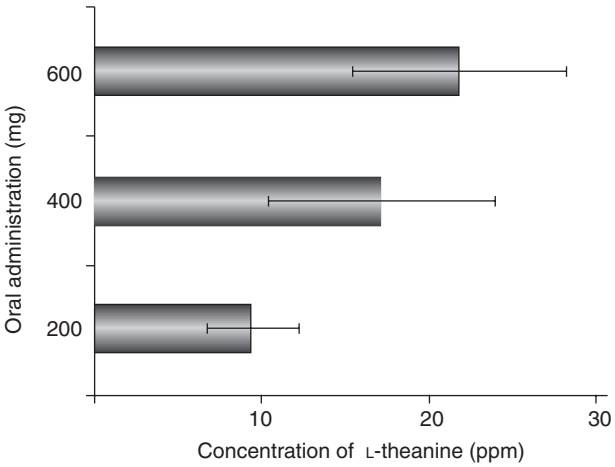

Figure 2. The relation between the plasma concentration of L-theanine and the dose of a single oral administration, and % prepulse inhibition (PPI). A significant positive correlation was revealed between the L-theanine plasma concentration and the single oral administration dose ($P < 0.001$). Error bar = SD. ppm, parts per million.

A significant positive correlation was revealed between the plasma concentration of L-theanine and the dose of oral administration ($P < 0.001$, Fig. 2). However, no significant correlation was observed between the plasma concentration of L-theanine and PPI metrics (data not shown).

DISCUSSION

We found that 200 mg or 400 mg of L-theanine increased the % PPI; however, 600 mg of L-theanine did not. Thus, administration of L-theanine significantly affected PPI of the subjects at a particular dose range. To our knowledge, this is the first study to obtain evidence of the effects of L-theanine on sensorimotor gating in humans.

The effects of L-theanine on PPI might be attributable to its chemical structure (i.e., as a glutamate derivative), and it is known that L-theanine has weak affinities for kainate, α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA), and NMDA receptors.[3,11] PPI is disrupted by the NMDA receptor antagonists phencyclidine and MK801.[32] However, glycine, which functions as a coagonist at the NMDA receptor, impaired the PPI in healthy humans.[33]

Different NMDA receptor antagonists, ketamine and memantine have been shown to increase or decrease PPI in healthy humans.[20,23,34] In ketamine, it was also reported that the changes of PPI treated with ketamine varied depending on the timing and dose of ketamine[35–38] The effect of ketamine on glutamate pathways is now thought to be more complex and strictly dose-dependent. A previous study demonstrated that various low doses of ketamine caused increases in glutamatergic outflow from prefrontal cortex in rats.[39] It is possible therefore, that cortical glutamatergic afferents can increase or decrease sensorimotor gating of information to the cortex, depending on baseline glutamatergic tone. In keeping with this notion, the other study suggests that glutamate can act either as a brake or as an accelerator of dopamine, serotonin, and noradrenaline systems, resulting in either increased or decreased information flowing to the cortex.[40] If a low dose of ketamine paradoxically acts as glutamate releasers then this would enhance the brake and could be seen to strengthen sensorimotor gating.[34] Thus, a narrow dose range for L-theanine-induced changes in PPI may result from the ambivalent action of glutamatergic transmission.

We observed the effect with 200–400 mg of L-theanine on % PPI, and this finding is in accordance with a previous study using memantine, a DA agonist as well as an NMDA antagonist.[41] Previous studies showed that a low dose of memantine increased PPI, whereas a high dose reduced PPI compared to controls in both rats and humans.[23] These points may indicate that the ratio of the dopamine agonist to NMDA antagonist effects of memantine differed at the doses they administered: a predominant NMDA antagonism at a low dose of memantine leads to increased PPI, and more prominent dopamine agonist effects at a high dose oppose these NMDA blockade effects.[23] L-theanine is known to have affinities for glutamatergic receptors,[3,11] and also to increase brain gamma-aminobutyric acid (GABA), serotonin, and dopamine levels.[8–10] However, comparative effects of L-theanine on the dopaminergic and glutamatergic systems are unknown. The NMDA receptor antagonist-induced disruptions of PPI are not reversed by haloperidol, indicating that these effects are not mediated only by the activation of central dopamine systems.[42] Thus, a narrow dose range for L-theanine-induced changes in PPI might also reflect its simultaneous action within multiple brain systems.

Our present results of the human study are inconsistent with our previous results that could not detect the effect of L-theanine on PPI in normal mice.[25] Some previous studies also showed a discrepancy between rodents and humans using ketamine, and

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amantadine,[20–23,35] which may be ascribed to genetic difference or administration dose.

The present study has some limitations. First, we found no correlation between the concentration of L-theanine and the PPI metrics. Our prior results showed that L-theanine had an effect with particular dose on % PPI, so it may be unfit to evaluate by a linear correlation analysis. It may also be suggested that PPI metrics vary greatly among individuals, and the interpersonal differences may be larger than the effect of L-theanine. Second, this study contained a relatively small sample size ($n = 14$), and two of the 14 subjects were smokers. It is well known that smoking influences the % PPI.[43,44] However, the subjects were forced to refrain from smoking for at least 20 min prior to the test. Further, to evaluate the smoking effect on the influence of L-theanine on the % PPI, we ran repeated measures Quade's rank analysis of covariance. However, there were no interactions between the smoking habit and % PPI in this study (data not shown). Additionally, some of these participants in this study were recruited from acquaintances of the authors. However, PPI is an objective test to measure the automatic response (i.e., startle reflex), so the potential bias due to participant selection is likely to be minimal. Despite these limitations, our results showed the effect with 200–400 mg of L-theanine on sensorimotor gating. Our findings suggest that L-theanine exerts its effects, at least in part, through an ambivalent action of the glutaminergic system. Further studies in schizophrenic patients, who were known to have deficient PPI, are warranted. In particular, studies investigating effect of chronic administration of L-theanine would provide more insight into the clinically significant benefits of L-theanine.

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